

Module 2: Transport Layer and Network Layer



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Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
 - select()
 - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using ip

Introduction to the Network Layer

- Communication between Alice and Bob occurs through the Network Layer.
- Same communication scenario used previously for:
 - Application Layer (Chapter 2)
 - Transport Layer (Chapter 3)
- Network Layer enables communication across interconnected networks.
- Responsible for packet delivery from source to destination.

Internet as an Internetwork

- Internet consists of many interconnected networks.
- Networks may be:
 - LANs (Local Area Networks)
 - WANs (Wide Area Networks)
- Networks are connected through:
 - Routers
 - Switches
- Internet is an **internetwork**:
 - Combination of LANs and WANs

Role of Connecting Devices

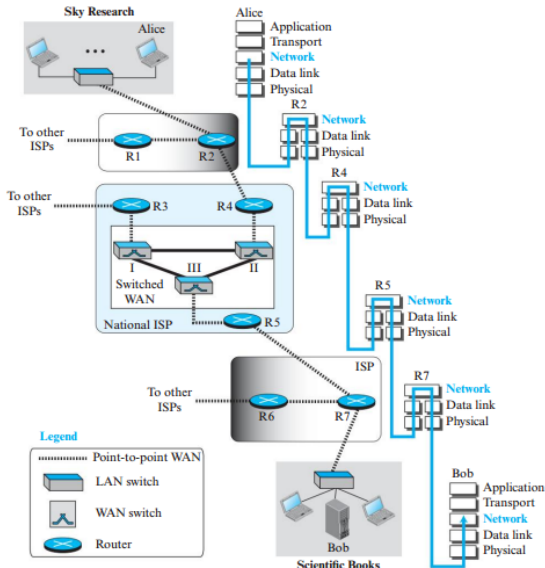
- Understanding the Network Layer requires understanding connecting devices.
- Routers and switches interconnect different networks.
- Enable packets to travel across multiple networks.
- Provide the communication path between source and destination hosts.
- Essential components of internetwork communication.

Network Layer Involvement

- Network Layer operates at:
 - Source Host (Alice)
 - Destination Host (Bob)
 - Intermediate Routers
- Example routers along the path:
 - R2
 - R4
 - R5
 - R7
- Every packet passes through the Network Layer at each point.

Communication at the network layer

Figure 4.1 Communication at the network layer



Functions at the Source Host

- Receives a packet from the Transport Layer.
- Encapsulates the packet into a datagram.
- Delivers the datagram to the Data-Link Layer.
- Data-Link Layer prepares the datagram for transmission.

Functions at the Destination Host

- Receives the datagram from the Data-Link Layer.
- Performs decapsulation.
- Extracts the packet from the datagram.
- Delivers the packet to the corresponding Transport Layer.
- Transport Layer continues further processing.

TCP/IP Layers Used by Hosts

- Source and destination hosts use all five TCP/IP layers:
 - ① Application Layer
 - ② Transport Layer
 - ③ Network Layer
 - ④ Data-Link Layer
 - ⑤ Physical Layer
- Hosts are responsible for end-to-end communication.

TCP/IP Layers Used by Routers

- Routers normally use only three layers for packet forwarding:
 - Network Layer
 - Data-Link Layer
 - Physical Layer
- Perform routing and forwarding functions.
- May use Transport and Application Layers for control purposes.

Router Architecture

- A router connects two or more networks.
- Receives a packet from one network.
- Forwards the packet to another network.
- Therefore, a router contains:
 - Two Data-Link Layers
 - Two Physical Layers
- One set handles incoming traffic.
- One set handles outgoing traffic.

Network-Layer Services

- Network layer provides several important services.
- Services expected from a network-layer protocol:
 - 1 Packetizing
 - 2 Routing
 - 3 Forwarding
 - 4 Error Control
 - 5 Flow Control
 - 6 Congestion Control
 - 7 Quality of Service (QoS)
 - 8 Security



- First duty of the network layer.
- Encapsulates payload received from upper layers.
- Creates a network-layer packet (datagram).
- Destination host decapsulates the datagram.
- Extracted payload is delivered to the upper layer.
- Payload should remain unchanged during transmission.

Packetizing at the Source Host

- Receives payload from an upper-layer protocol.
- Adds a network-layer header.
- Header contains:
 - Source address
 - Destination address
 - Other protocol information
- Creates a datagram.
- Delivers datagram to the data-link layer.
- Payload cannot be changed unless fragmentation is required.

Packetizing at the Destination Host

- Receives datagram from the data-link layer.
- Removes the network-layer header.
- Extracts the payload.
- Delivers payload to the corresponding upper-layer protocol.
- Responsible for reassembly of fragmented packets.
- Waits until all fragments arrive before delivery.

Packetizing at Routers

- Routers do not normally decapsulate packets.
- Routers cannot modify:
 - Source address
 - Destination address
- Routers inspect addresses only for forwarding.
- Fragmentation may occur when necessary.
- Header is copied to each fragment.
- Additional header modifications may be required.

- Second major responsibility of the network layer.
- Determines path from source to destination.
- Internet consists of:
 - Multiple LANs
 - Multiple WANs
 - Interconnecting routers
- Multiple routes may exist between source and destination.
- Network layer selects the best route.



Routing Protocols

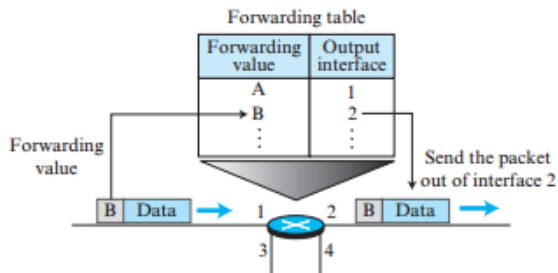
- Routers require knowledge of network topology.
- Routing protocols help routers:
 - Exchange information
 - Learn about neighbors
 - Build consistent routing tables
- Routing protocols run before communication begins.
- Routing tables are used during packet delivery.

Forwarding

- Forwarding is different from routing.
- Routing creates decision-making tables.
- Forwarding uses these tables.
- Occurs whenever a packet arrives at a router interface.
- Router decides where to send the packet next.

Forwarding

Figure 4.2 *Forwarding process*



Forwarding Process

- Router receives packet from an attached network.
- Uses information in packet header:
 - Destination address
 - Label
- Searches forwarding table (routing table).
- Determines output interface number.
- Sends packet to:
 - One network (Unicast)
 - Multiple networks (Multicast)

- Error control exists in:
 - Transport Layer
 - Data-Link Layer
- Internet network layer does not provide full error control.
- Fragmentation at routers makes error checking inefficient.
- Data payload is not checked by the network layer.

Header Checksum

- Network layer includes a checksum field.
- Checksum protects only the header.
- Entire datagram is not checked.
- Detects corruption in header information.
- Helps preserve correctness of routing information.

ICMP and Error Reporting

- Internet uses ICMP as an auxiliary protocol.
- ICMP provides limited error reporting.
- Used when:
 - Datagram is discarded
 - Unknown header information is detected
- Complements the network layer.

- Flow control regulates transmission rate.
- Prevents sender from overwhelming receiver.
- Receiver may need to send feedback to sender.
- Feedback indicates inability to process data quickly.

Why Internet Network Layer Lacks Flow Control

- Datagrams are sent whenever they are ready.
- No direct flow control mechanism exists.
- Reasons:
 - 1 Receiver-side network-layer processing is simple.
 - 2 Upper layers can buffer incoming data.
 - 3 Upper-layer protocols already provide flow control.
 - 4 Additional flow control increases complexity.

Congestion Control

- Congestion occurs when too many datagrams exist in the network.
- Network or router capacity may be exceeded.
- Routers may begin dropping packets.
- Lost packets can trigger retransmissions.
- Retransmissions may worsen congestion.

Effects of Congestion

- Increased packet loss.
- Increased retransmissions.
- Reduced network performance.
- Possible congestion collapse.
- In extreme cases:
 - No datagrams are successfully delivered.
- Discussed conceptually but not implemented in Internet network layer.

Quality of Service (QoS)

- QoS has become important due to multimedia applications.
- Examples:
 - Audio streaming
 - Video streaming
 - Real-time communication
- Internet has evolved to provide better QoS support.
- Most QoS mechanisms are implemented in upper layers.

- Security was not a concern in the early Internet.
- Original Internet users were mainly researchers and universities.
- Network layer initially included no security mechanisms.
- Modern Internet requires strong security protection.

IPSec and Network Security

- Security requires additional mechanisms.
- IPSec provides security at the network layer.
- IPSec creates a virtual security layer.
- Converts connectionless service into a secure connection-oriented service.
- Provides protection for modern Internet communications.

Packet Switching

- Routing and forwarding imply switching at the Network Layer.
- A router acts as a switch.
- Connects an input port to an output port (or multiple output ports).
- Similar to an electrical switch that allows current to flow.
- Network Layer uses **Packet Switching**.
- Circuit Switching is mainly used at the Physical Layer.

Packet-Switched Networks

- Message from upper layer is divided into packets.
- Packets are sent independently through the network.
- Source sends packets one by one.
- Destination receives packets one by one.
- Destination waits for all packets before reconstructing the message.
- Switching devices determine routes to the destination.

Packet-Switching Approaches

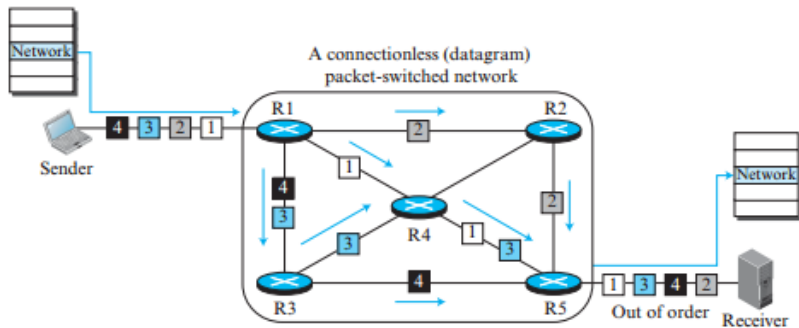
- Two approaches are used in packet-switched networks:
 - ① Datagram Approach
 - ② Virtual-Circuit Approach
- Both approaches perform routing and forwarding.
- Differ in how packets are handled inside the network.

Datagram Approach

- Also called **Connectionless Service**.
- Original Internet design adopted this approach.
- Each packet is treated independently.
- No relationship exists between packets of the same message.
- Network Layer only provides packet delivery.

A connectionless packet-switched network

Figure 4.3 A connectionless packet-switched network



Characteristics of Datagram Service

- Packets may follow different paths.
- Packets may arrive out of order.
- No setup phase is required.
- No dedicated path exists between source and destination.
- Routers operate independently for each packet.

Datagram Packet Routing

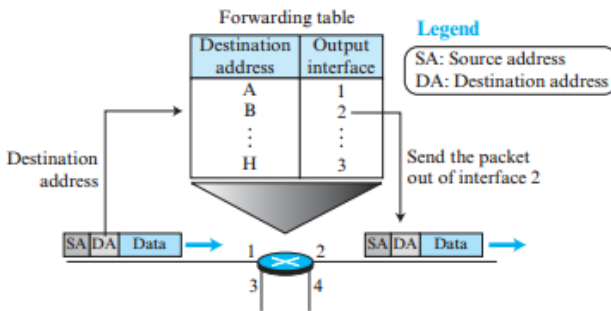
- Each packet contains:
 - Source Address
 - Destination Address
- Destination address determines where the packet goes.
- Source address identifies where the packet originated.
- Router forwards packets using only the destination address.
- Source address may be used for error reporting.

Datagram Service Summary

- No connection establishment.
- Each packet is an independent entity.
- Packets from different messages may intermingle.
- Flexible routing.
- Core forwarding decision based on destination address.

Forwarding process in a router when used in a connectionless network

Figure 4.4 Forwarding process in a router when used in a connectionless network

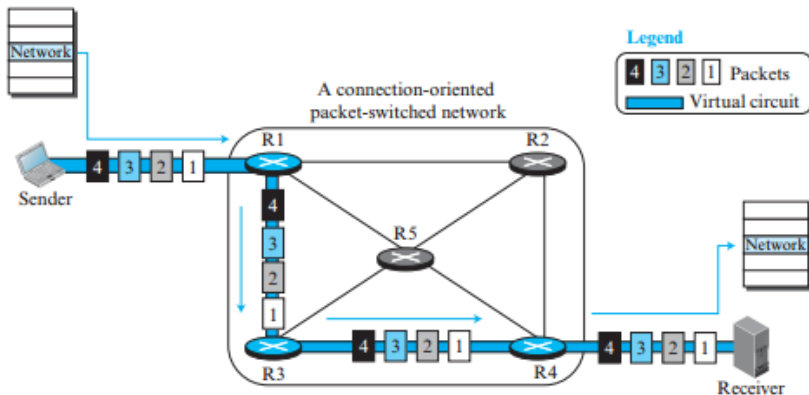


Virtual-Circuit Approach

- Also called **Connection-Oriented Service**.
- All packets belonging to a message are related.
- A virtual connection is established before data transfer.
- All packets follow the same predefined path.
- Provides ordered delivery of packets.

A virtual-circuit packet-switched network

Figure 4.5 A virtual-circuit packet-switched network



Virtual Circuit Identifier

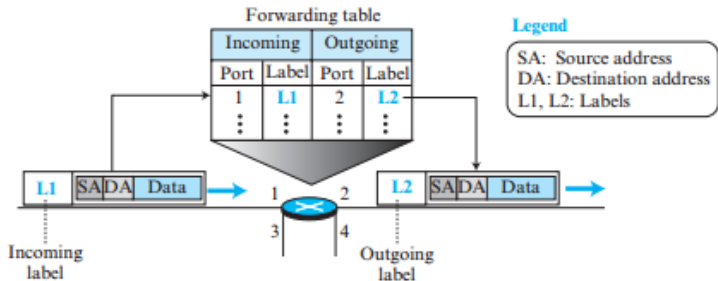
- Each packet carries:
 - Source Address
 - Destination Address
 - Flow Label
- Flow Label = Virtual Circuit Identifier (VCI).
- Defines the path followed by packets.
- Forwarding decisions are based on the label.

Forwarding in Virtual Circuits

- Router examines packet label.
- Label identifies the virtual path.
- Router uses forwarding table entries.
- Packet is forwarded to the correct output interface.
- Forwarding decision depends on label value.

Forwarding process in a router when used in a virtual-circuit network

Figure 4.6 Forwarding process in a router when used in a virtual-circuit network



Three Phases of Virtual-Circuit Service

- Virtual-circuit communication consists of:

- ① Setup Phase
- ② Data-Transfer Phase
- ③ Teardown Phase

- Setup creates forwarding table entries.
- Data transfer uses established entries.
- Teardown removes entries.

Setup Phase

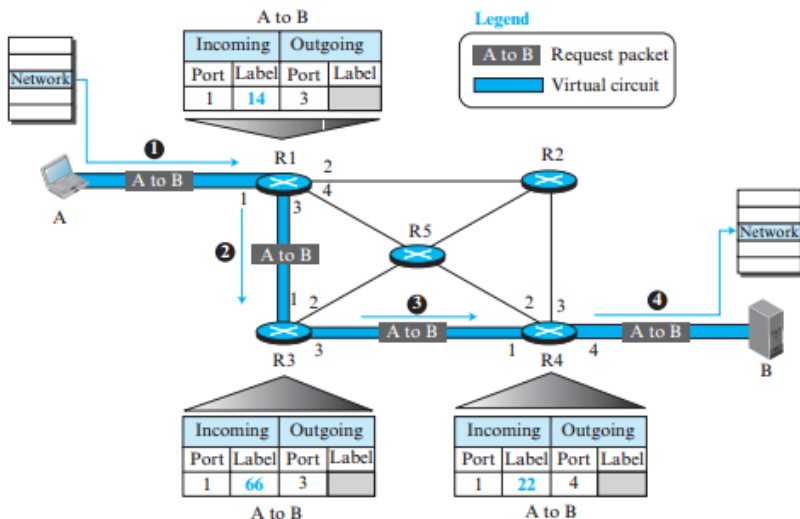
- Establishes the virtual circuit.
- Uses source and destination addresses.
- Creates forwarding table entries in routers.
- Requires exchange of:
 - Request Packet
 - Acknowledgment Packet

Request Packet

- Sent from source to destination.
- Carries:
 - Source Address
 - Destination Address
- Routers create partial virtual-circuit entries.
- Routers assign:
 - Incoming Port
 - Incoming Label
 - Outgoing Port

Sending request packet in a virtual-circuit network

Figure 4.7 Sending request packet in a virtual-circuit network



Request Packet Example

- Source A sends request to Router R1.
- R1 creates:
 - Incoming Port = 1
 - Incoming Label = 14
 - Outgoing Port = 3
- Packet forwarded to Router R3.
- Similar process occurs at R3 and R4.

Destination Processing

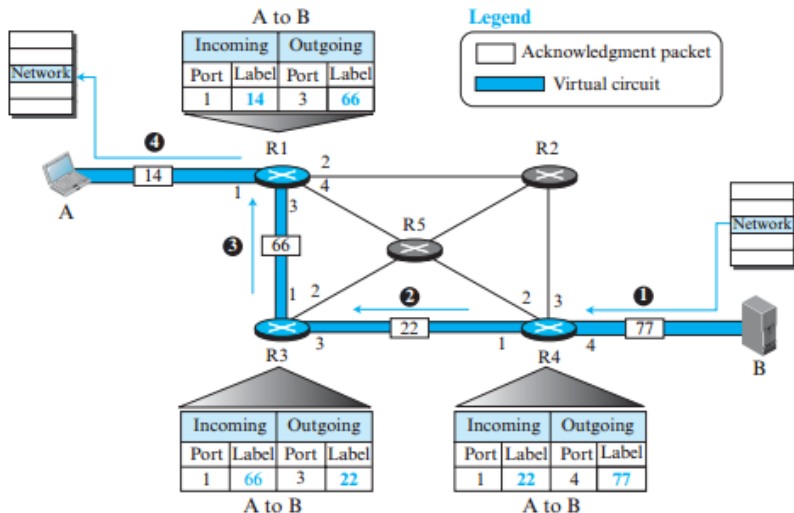
- Destination B receives setup packet.
- Assigns an incoming label.
- Example:
 - Incoming Label = 77
- Label identifies packets arriving from Source A.

Acknowledgment Packet

- Sent from destination back to source.
- Completes forwarding table entries.
- Carries:
 - Source Address
 - Destination Address
 - Assigned Label
- Routers fill the outgoing label field.

Sending acknowledgments in a virtual-circuit network

Figure 4.8 Sending acknowledgments in a virtual-circuit network



Acknowledgment Process

- Destination B sends acknowledgment with label 77.
- Router R4 sets outgoing label = 77.
- R4 sends acknowledgment to R3.
- R3 sets outgoing label.
- R3 forwards acknowledgment to R1.
- R1 completes its entry.
- Source A receives final label information.

Completed Virtual Circuit

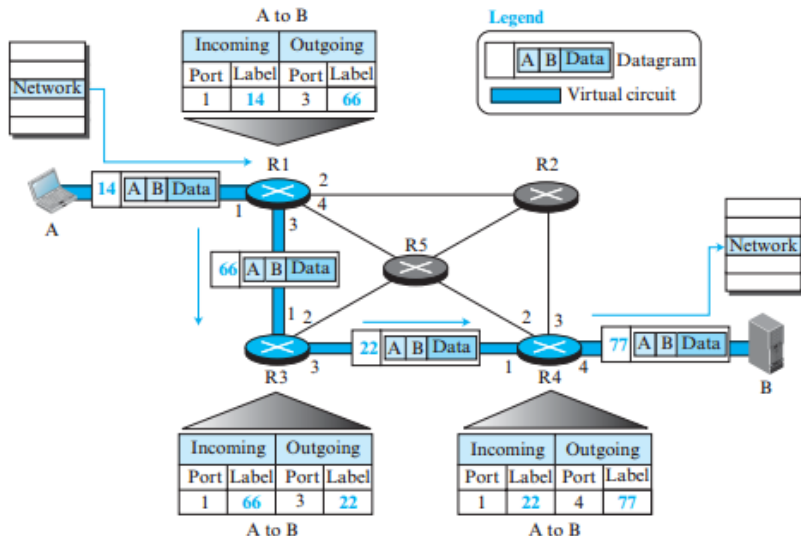
- After acknowledgment:
 - All routers have complete forwarding entries.
 - Virtual path is established.
- Communication can now begin.

Data-Transfer Phase

- Begins after setup completion.
- All packets follow the established virtual circuit.
- Routers use labels for forwarding.
- No route calculation is required during transfer.

Flow of one packet in an established virtual circuit

Figure 4.9 Flow of one packet in an established virtual circuit



Label Translation During Transfer

- Source A sends packet with Label 14.
- Router R1 changes label to 66.
- Router R3 changes label to 22.
- Router R4 changes label to 77.
- Destination B receives packet with Label 77.

Advantages of Virtual Circuits

- All packets follow the same path.
- Packets arrive in order.
- Faster forwarding using labels.
- Predictable route through the network.

Teardown Phase

- Begins after all packets are delivered.
- Source sends a teardown packet.
- Destination sends a confirmation packet.
- Routers delete virtual-circuit entries.
- Network resources are released.

Datagram vs Virtual Circuit

Feature	Datagram	Virtual Circuit
Service Type	Connectionless	Connection-Oriented
Setup Required	No	Yes
Routing Basis	Destination Address	Label (VCI)
Path	May Vary	Fixed
Packet Order	Not Guaranteed	Guaranteed
Forwarding	Per Packet	Per Label
Resource Reservation	No	Possible

Network-Layer Performance

- Network layer does not provide ideal service.
- Performance is measured using:
 - 1 Delay
 - 2 Throughput
 - 3 Packet Loss
- These factors directly affect communication quality.



- Packets experience delays while traveling through the network.
- Delay is the time taken for a packet to reach its destination.
- Four major types of delay:
 - 1 Transmission Delay
 - 2 Propagation Delay
 - 3 Processing Delay
 - 4 Queuing Delay

Transmission Delay

- Time required to place all packet bits onto the transmission medium.
- Sender transmits bits one by one.
- Depends on:
 - Packet Length
 - Transmission Rate


$$Delay_{tr} = \frac{\text{Packet Length}}{\text{Transmission Rate}}$$

Transmission Delay Example

- Fast Ethernet transmission rate:

$$100 \times 10^6 \text{ bps}$$

- Packet length:



$$Delay_{tr} = \frac{10,000}{100,000,000} = 100 \mu s$$

Propagation Delay

- Time required for a bit to travel through the medium.
- Depends on:
 - Distance of the link
 - Propagation speed of the medium
- Speed in vacuum:


$$3 \times 10^8 \text{ m/s}$$

$$Delay_{pg} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Propagation Delay Example

- Link distance:

2000 m

- Propagation speed:

$2 \times 10^8 \text{ m/s}$

$$Delay_{pg} = \frac{2000}{2 \times 10^8} = 10 \mu s$$

Processing Delay

- Time required by a router or destination host to process a packet.
- Includes:
 - Receiving packet
 - Removing header
 - Error checking
 - Forwarding or delivery
- Usually expressed as an average value.

$$Delay_{pr} = \text{Time required to process a packet}$$

Queuing Delay

- Time a packet waits in router queues.
- Routers contain:
 - Input Queues
 - Output Queues
- Occurs when packets arrive faster than they can be processed.
- Similar to airplanes waiting for landing or departure.

$$Delay_{qu} = \text{Waiting Time in Router Queues}$$

Total End-to-End Delay

- Assume:
 - n routers in the path
 - $n + 1$ links
- Total delay includes:
 - Transmission delays
 - Propagation delays
 - Processing delays
 - Queuing delays

$$Total\ Delay = (n + 1)(Delay_{tr} + Delay_{pg} + Delay_{pr}) + n(Delay_{qu})$$

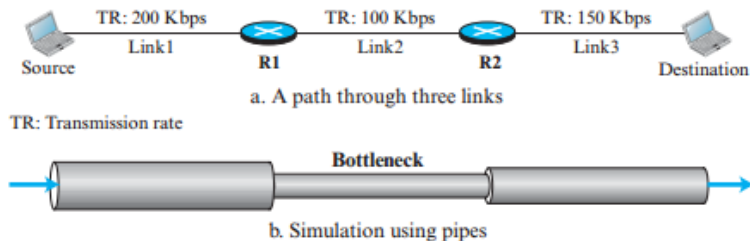
Throughput

- Throughput is the number of bits passing through a point per second.
- Represents the actual data transfer rate.
- Measured in:
 - bps
 - Kbps
 - Mbps
 - Gbps
- Different links may have different transmission rates.



Throughput in a path with three links in a series

Figure 4.10 *Throughput in a path with three links in a series*



Throughput in a Multi-Link Path

- Consider three links:
 - Link 1 = 200 Kbps
 - Link 2 = 100 Kbps
 - Link 3 = 150 Kbps
- Data flow is limited by the slowest link.
- Slowest link is called the **Bottleneck**.

$$\text{Throughput} = 100 \text{ Kbps}$$

General Throughput Formula

- For n serial links:
- End-to-end throughput equals the minimum link rate.
- Bottleneck determines overall performance.

$$\text{Throughput} = \min\{TR_1, TR_2, \dots, TR_n\}$$

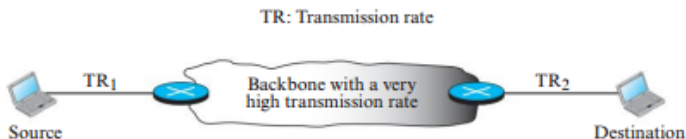
Throughput in the Internet

- Typical Internet path consists of:
 - Source Access Network
 - Internet Backbone
 - Destination Access Network
- Backbone speed is usually very high.
- Throughput is normally determined by access links.

$$Throughput = \min(TR_1, TR_2)$$

A path through the Internet backbone

Figure 4.11 *A path through the Internet backbone*



Internet Throughput Example

- Server connection:

100 Mbps

- User connection:

40 Kbps

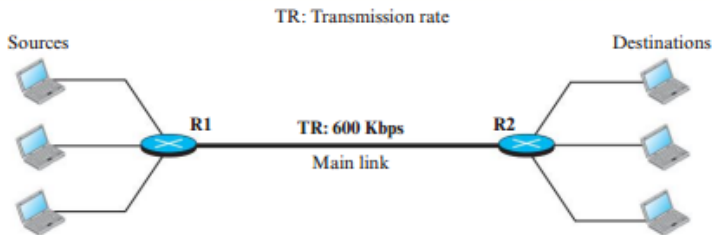
- Bottleneck:

- Dial-up connection

$\text{Throughput} = 40\text{ Kbps}$

Effect of throughput in shared links

Figure 4.12 *Effect of throughput in shared links*



Shared-Link Throughput

- Links between routers may be shared.
- Multiple flows use the same link.
- Available bandwidth is divided among flows.
- Shared capacity affects throughput calculations.
- Example:
 - Main link shared by three paths.
 - Effective rate per path decreases.

- Packet loss is another important performance metric.
- Occurs when routers cannot store incoming packets.
- Routers use finite-sized input buffers.
- Buffer overflow causes packet drops.

Causes of Packet Loss

- Router is busy processing another packet.
- Incoming packets wait in buffers.
- Buffer becomes full.
- New packets are discarded.
- Common during congestion.

Effects of Packet Loss

- Lost packets must be retransmitted.
- Retransmissions increase network traffic.
- Increased traffic may create further congestion.
- Additional congestion may cause more packet loss.
- Can significantly reduce network performance.

Summary of Network-Layer Performance

Metric	Description
Delay	Time taken by a packet to reach destination
Transmission Delay	Time to place bits on the medium
Propagation Delay	Time for bits to travel through the medium
Processing Delay	Time to process a packet
Queuing Delay	Time waiting in router queues
Throughput	Actual data transfer rate
Packet Loss	Packets discarded due to buffer overflow

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- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

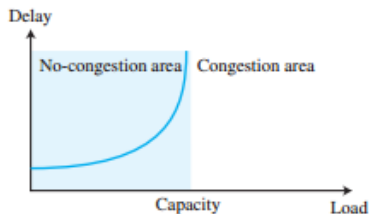
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Network-Layer Congestion

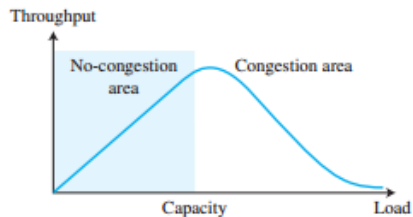
- Congestion at the Network Layer is related to:
 - Throughput
 - Delay
- Not explicitly addressed in the Internet model.
- Helps us understand congestion at the Transport Layer.
- Can suggest remedies for network-layer congestion.

Packet delay and throughput as functions of load

Figure 4.13 *Packet delay and throughput as functions of load*



a. Delay as a function of load



b. Throughput as a function of load

Load vs Delay

- Delay remains minimal when load is below network capacity.
- Minimum delay consists of:
 - Propagation Delay
 - Processing Delay
- As load approaches network capacity:
 - Queuing Delay increases.
 - Total Delay rises sharply.
- Delay becomes theoretically infinite when load exceeds capacity.

Load vs Throughput

- Throughput increases proportionally with load initially.
- Maximum throughput occurs near network capacity.
- Beyond capacity:
 - Routers start dropping packets.
 - Throughput decreases sharply.
- Packet retransmissions increase network load.
- Congestion worsens instead of improving.

Why Throughput Decreases During Congestion

- Router queues become full.
- Incoming packets are discarded.
- Sources detect packet loss.
- Lost packets are retransmitted.
- Retransmissions create additional traffic.
- Additional traffic intensifies congestion.

Congestion Control

- Congestion control aims to:
 - Prevent congestion before it occurs.
 - Remove congestion after it occurs.
- Two broad categories:
 - ① Open-Loop Congestion Control (Prevention)
 - ② Closed-Loop Congestion Control (Removal)

Open-Loop Congestion Control

- Preventive approach.
- Policies are applied before congestion occurs.
- Congestion control handled by:
 - Source
 - Destination
- Objective:
 - Avoid congestion proactively.

Retransmission Policy

- Retransmission may be necessary when packets are lost.
- Excessive retransmissions increase congestion.
- Efficient retransmission policy should:
 - Detect losses accurately.
 - Avoid unnecessary retransmissions.
 - Optimize retransmission timers.
- Goal:
 - High efficiency with low congestion.

- Sender window mechanism affects congestion.
- Selective Repeat is better than Go-Back-N.

Go-Back-N	Selective Repeat
Resends multiple packets	Resends only lost packets
Creates duplicates	Minimizes duplicates
May worsen congestion	Better congestion control

Acknowledgment Policy

- Acknowledgments contribute to network load.
- Fewer acknowledgments reduce congestion.
- Possible strategies:
 - Delayed acknowledgments.
 - Piggyback acknowledgments.
 - Acknowledge every N packets.
 - Send acknowledgment after timer expiration.

Discarding Policy

- Routers may discard packets during congestion.
- Intelligent packet dropping can reduce congestion.
- Example:
 - Audio applications discard less important packets.
- Maintains acceptable quality while reducing load.

- Used in Virtual-Circuit Networks.
- Related to Quality of Service (QoS).
- Router checks resource requirements before admission.
- Connection may be denied if:
 - Congestion already exists.
 - Future congestion is likely.

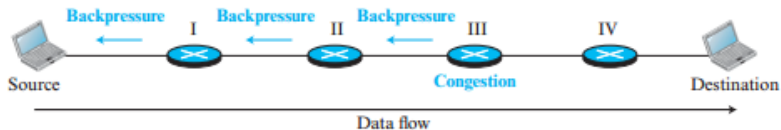
Closed-Loop Congestion Control

- Reactive approach.
- Takes action after congestion occurs.
- Detects congestion.
- Applies corrective measures.
- Common mechanisms:
 - Backpressure
 - Choke Packet
 - Implicit Signaling
 - Explicit Signaling

- Congested node stops accepting data.
- Requests upstream node to slow down.
- Congestion notification propagates backward.
- Eventually reaches the source.
- Node-to-node congestion control mechanism.

Backpressure method for alleviating congestion

Figure 4.14 *Backpressure method for alleviating congestion*



Backpressure Operation

- 1 Node III becomes congested.
- 2 Node III asks Node II to slow down.
- 3 Node II may become congested.
- 4 Node II asks Node I to slow down.
- 5 Node I informs the source.
- 6 Source reduces transmission rate.

Result: Congestion pressure moves backward toward the source.

Limitation of Backpressure

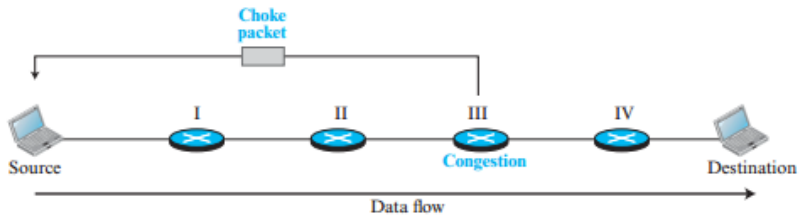
- Works only in Virtual-Circuit Networks.
- Each node must know its upstream node.
- Cannot be used in Datagram Networks.
- Routers in datagram networks do not track upstream paths.

Choke Packet

- Router directly informs the source about congestion.
- Uses a special packet called a choke packet.
- Intermediate routers take no action.
- Warning travels directly to the source.

Choke Packets

Figure 4.15 Choke packet



Backpressure vs Choke Packet

Backpressure	Choke Packet
Node-to-node warning	Router-to-source warning
Moves hop-by-hop	Direct notification
Only virtual circuits	Can be used more generally
Gradual propagation	Immediate warning

Source Quench in ICMP

- Internet uses ICMP for congestion notifications.
- Router may discard overloaded datagrams.
- Router sends:
 - Source Quench ICMP Message
- Source reduces transmission rate.
- Intermediate routers do not participate.

Implicit Signaling

- No explicit communication about congestion.
- Source infers congestion indirectly.
- Indicators include:
 - Missing acknowledgments.
 - Increased delays.
 - Frequent timeouts.
- Source automatically slows down transmission.

Implicit Signaling Example

- Source sends packets.
- Acknowledgments are delayed.
- Source assumes network congestion.
- Transmission rate is reduced.
- Used in TCP Congestion Control.

Explicit Signaling

- Congested node explicitly signals congestion.
- No separate control packet is required.
- Congestion information is embedded inside data packets.
- Can operate in:
 - Forward Direction
 - Backward Direction

Explicit Signaling vs Choke Packet

Choke Packet	Explicit Signaling
Uses separate packet	Uses existing packets
Additional network traffic	No extra packet overhead
Direct warning to source	Signal embedded in data flow

Summary of Congestion Control

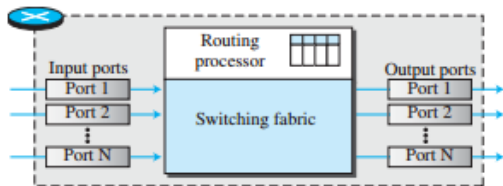
Technique	Purpose
Retransmission Policy	Optimize retransmissions
Window Policy	Reduce duplicate packets
Acknowledgment Policy	Reduce ACK traffic
Discarding Policy	Intelligent packet dropping
Admission Policy	Prevent overload
Backpressure	Hop-by-hop slowdown
Choke Packet	Direct source notification
Implicit Signaling	Infer congestion indirectly
Explicit Signaling	Embed congestion information

Structure of a Router

- A router is often represented as a black box.
- Receives packets through input ports.
- Uses a forwarding table to determine output ports.
- Sends packets through output ports.
- Internally, a router consists of several functional components.

Router components

Figure 4.16 *Router components*



Major Components of a Router

- A router consists of four main components:
 - 1 Input Ports
 - 2 Output Ports
 - 3 Routing Processor
 - 4 Switching Fabric
- These components work together to forward packets efficiently.

Input Ports

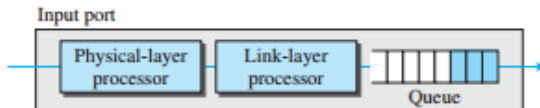
- Perform Physical Layer functions.
- Perform Data-Link Layer functions.
- Receive signals from the transmission medium.
- Convert signals into bits.
- Extract packets from incoming frames.

Functions of an Input Port

- Reconstruct bits from received signals.
- Decapsulate packets from frames.
- Check frames for errors.
- Discard corrupted packets.
- Prepare packets for Network Layer processing.

Input Port

Figure 4.17 *Input port*



Input Port Architecture

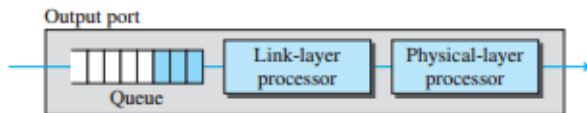
- Consists of:
 - Physical-Layer Processor
 - Data-Link-Layer Processor
 - Input Buffers (Queues)
- Buffers temporarily store packets.
- Packets wait before entering the switching fabric.

Output Ports

- Perform operations opposite to input ports.
- Prepare packets for transmission.
- Handle Physical Layer and Data-Link Layer tasks.
- Send packets to the outgoing communication link.

Output Port

Figure 4.18 *Output port*



Functions of an Output Port

- Store outgoing packets in queues.
- Encapsulate packets into frames.
- Apply Physical Layer processing.
- Generate signals for transmission.
- Send signals onto the communication medium.

Output Port Architecture

- Consists of:
 - Output Queues
 - Data-Link-Layer Processor
 - Physical-Layer Processor
- Queues packets before transmission.
- Ensures orderly packet delivery.

Routing Processor

- Performs Network Layer functions.
- Responsible for routing and forwarding decisions.
- Examines destination addresses.
- Determines:
 - Next Hop
 - Output Port Number

Routing Table Lookup

- Routing processor searches the forwarding table.
- Process is called:
 - Table Lookup
- Maps destination addresses to output ports.
- Essential for packet forwarding.

Modern Router Optimization

- Traditional routers perform lookup in the routing processor.
- Modern routers move lookup functions to input ports.
- Benefits:
 - Faster forwarding
 - Reduced processing delay
 - Higher throughput

Switching Fabric

- Connects input ports to output ports.
- Moves packets through the router.
- Most critical component for router performance.
- Switching speed affects:
 - Queue sizes
 - Packet delay
 - Throughput

Evolution of Switching Fabric

- Early routers were dedicated computers.
- Packet transfer used:
 - Shared Memory
 - Shared Bus
- Input ports stored packets in memory.
- Output ports retrieved packets from memory.
- Modern routers use advanced switching fabrics.

Types of Switching Fabrics

- Common switching fabrics include:

- 1 Crossbar Switch

- 2 Banyan Switch

- 3 Batcher-Banyan Switch

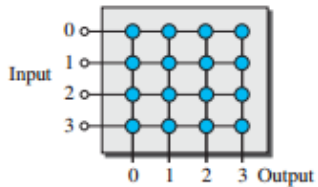
Crossbar Switch

- Simplest switching fabric.
- Connects:
 - n Inputs
 - n Outputs
- Uses electronic microswitches.
- Organized as a grid structure.
- Each intersection contains a crosspoint switch.



Crossbar Switch

Figure 4.19 *Crossbar switch*



Crossbar Switch Characteristics

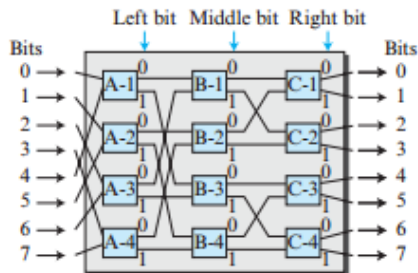
- Direct connection between inputs and outputs.
- High switching speed.
- Simple architecture.
- Hardware complexity increases rapidly as n increases.

Banyan Switch

- More realistic than a crossbar switch.
- Multistage switching network.
- Uses microswitches at each stage.
- Routes packets based on output-port binary addresses.

Banyan Switch

Figure 4.20 Banyan switch

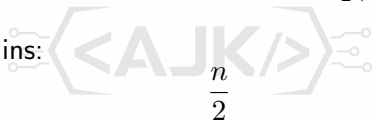


Banyan Switch Structure

- For n inputs and n outputs:

$$\text{Number of Stages} = \log_2(n)$$

- Each stage contains:



microswitches.

- Each stage examines one bit of the destination address.

Banyan Switch Routing Process

- First stage examines highest-order bit.
- Second stage examines second-highest-order bit.
- Process continues stage by stage.
- Final stage determines exact output port.

Banyan Switch Example 1

- Input Port = 1
- Output Port = 6
- $6 = 110_2$
- Stage 1 routes using bit 1.
- Stage 2 routes using bit 1.
- Stage 3 routes using bit 0.
- Packet reaches Output Port 6.

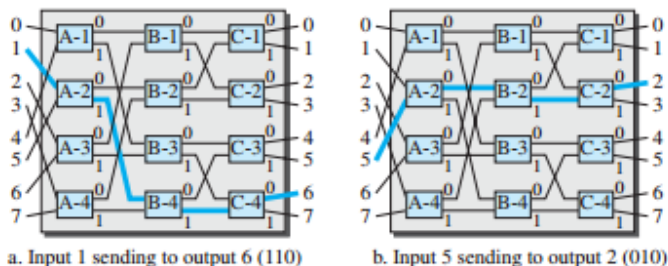
Banyan Switch Example 2

- Input Port = 5
- Output Port = 2
- Stage 1 routes using bit 0.
- Stage 2 routes using bit 1.
- Stage 3 routes using bit 0.
- Packet reaches Output Port 2.

$$2 = 010_2$$

Examples of Banyan Switch

Figure 4.21 *Examples of routing in a banyan switch*



Limitation of Banyan Switch

- Internal collisions may occur.
- Collision possible even when packets have different destinations.
- Simultaneous packets may compete for internal paths.
- Reduces switching efficiency.

Batcher-Banyan Switch

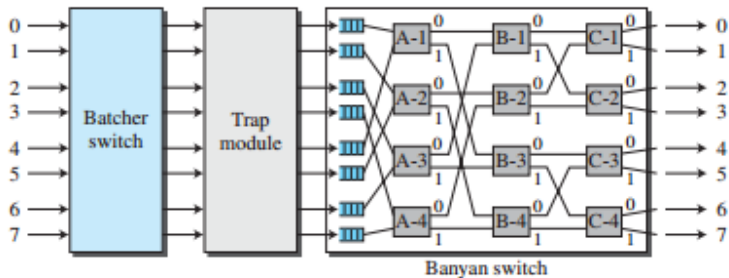
- Designed to overcome Banyan switch limitations.
- Combines:
 - Batcher Sorting Switch
 - Banyan Switch
- Incoming packets are sorted before switching.

Batcher Sorting Switch

- Developed by K. E. Batcher.
- Sorts incoming packets.
- Sorting is based on destination addresses.
- Reduces internal switching conflicts.
- Improves overall router performance.

Batcher Sorting Switch

Figure 4.22 Batcher-banyan switch



- Added between:
 - Batchter Switch
 - Banyan Switch
- Prevents duplicate destination conflicts.
- Ensures only one packet per destination proceeds per clock tick.

Operation of the Trap Module

- Multiple packets may target the same destination.
- Trap module allows only one packet to pass.
- Remaining packets wait for the next clock tick.
- Eliminates simultaneous destination conflicts.

Comparison of Switching Fabrics

Switch Type	Structure	Main Feature
Crossbar	Single Stage Grid	Direct Input-Output Connections
Banyan	Multi-Stage	Routing Based on Binary Address Bits
Batcher-Banyan	Sorting + Banyan	Eliminates Internal Collisions

Summary: Structure of a Router

- Router consists of:
 - ① Input Ports
 - ② Output Ports
 - ③ Routing Processor
 - ④ Switching Fabric
- Switching fabric determines forwarding efficiency.
- Modern routers use advanced switching architectures.
- Batcher-Banyan switch provides efficient collision-free routing.